

Lightweight Multimodal Person Detection and Temporal Post-Processing for Aerial Search and Rescue

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Abstract—Autonomous unmanned aerial vehicles (UAVs) deployed in wilderness search-and-rescue (SAR) missions require reliable person detection under severe size, weight, power, and computing (SWaP-C) constraints. While multimodal sensing combining visible red-green-blue (RGB) and thermal infrared (TIR) cameras mitigates single-modality dropouts caused by canopy shade, darkness, or thermal crossover, frame-level object detectors frequently produce transient false positive spikes. In autonomous aerial operations, each false alarm triggers wasteful loiter maneuvers, battery depletion, and spurious transmissions over bandwidth-constrained Internet of Things (IoT) satellite or cellular links. This paper presents a systematic comparative evaluation of lightweight causal temporal decision methods designed to generate reliable and efficient alerts from multimodal visible RGB and thermal detections for autonomous aerial search and rescue. We freeze our study around two wilderness environments (arid desert and temperate forest) and two visual modalities (visible RGB and thermal infrared). We deploy two upstream lightweight You Only Look Once (YOLO11n) detectors (2.6 million parameters, 6.5 giga-floating-point operations (GFLOPs) at 640×640 resolution): one trained on multi-environment RGB sequences and one on multi-environment thermal sequences, using batch size 16 in single-precision floating-point (FP32) arithmetic for numerical stability on embedded avionics. Frame confidences are integrated via soft disjunctive late fusion and filtered across a causal five-frame sliding window (208 ms latency bound at 24 Hz). We evaluate five temporal decision strategies: baseline raw thresholding, five-frame moving average, five-frame median filtering, five-frame history consensus, and a learned pure-PyTorch Mamba selective state-space model (SSSM), with decision thresholds optimized systematically on validation sequences. Methods are benchmarked across four core dimensions: positive event detection, false alert mitigation, decision delay, and edge processing overhead.

Index Terms—Multimodal fusion, search and rescue, unmanned aerial vehicles, autonomous alert generation, state-space models, lightweight object detection.

I. INTRODUCTION

Autonomous unmanned aerial vehicles (UAVs) have become critical assets in wilderness search-and-rescue (SAR)

operations [1], [2]. In vast or inaccessible terrains such as arid deserts and dense forests, aerial robotic platforms survey expansive disaster sectors far faster than ground search crews. However, edge-deployed computer vision systems operate under severe size, weight, power, and computing (SWaP-C) constraints. Furthermore, remote SAR missions lack high-bandwidth communication infrastructure. Autonomous UAVs must transmit alarm telemetry over low-bandwidth cellular or satellite Internet of Things (IoT) uplinks [3]. Under these conditions, an alarm triggers immediate physical consequences: the drone interrupts its search trajectory to loiter and verify the target, depleting onboard battery reserves, or dispatches rescue crews across hazardous ground. Consequently, false alarms carry severe operational penalties.

Optical sensing in wilderness environments is susceptible to modality-specific failures. Standard red-green-blue (RGB) cameras fail in dense canopy shadows, fog, and darkness. Conversely, thermal infrared (TIR) cameras detect emitted human body heat regardless of illumination [4], [5]. Nevertheless, thermal sensors encounter thermal crossover, where human skin temperatures equilibrate with sun-heated sand, rocks, or asphalt [6]. Sun-warmed ground objects, rocky outcroppings, and wildlife frequently produce thermal anomalies that confuse visual detectors. These operational realities mandate dual-spectrum sensing where visible and thermal streams provide complementary evidence.

To satisfy real-time constraints on edge processors, aerial platforms rely on compact single-stage detectors such as YOLO [7], [8]. In this work, we deploy Ultralytics YOLO11n [9] (2.6 million parameters, 6.5 GFLOPs at 640×640 resolution). To prevent half-precision numerical instability on embedded GPUs, we train the detector with batch size 16 in single-precision floating-point (FP32) arithmetic. Although lightweight models achieve high inference throughput, frame-level detections exhibit temporal volatility. Confidence scores oscillate due to aerial platform vibration, foliage occlusions, viewpoint changes, and fleeting distractors. Direct thresholding of raw detections causes isolated confidence spikes to trigger

The author(s) confirm(s) that all human/animal subject research procedures and protocols are exempt from review board approval.

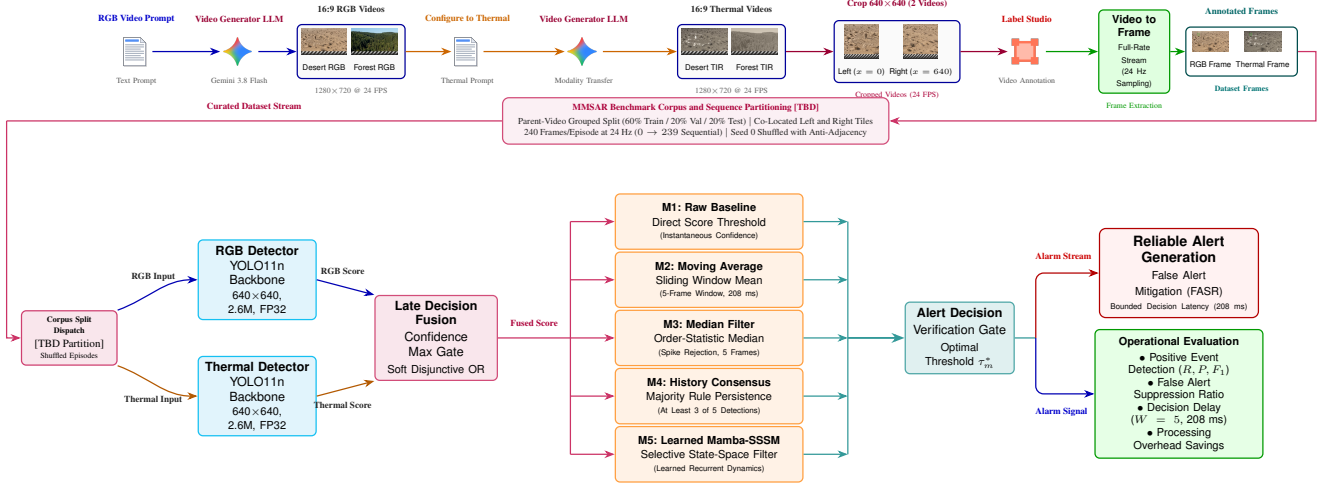


Fig. 1. System architecture of the proposed Multimodal Search and Rescue (MMSAR) evaluation framework. Visible RGB and thermal infrared video streams are synthesized from text prompts and Gemini video generation, followed by cross-modality thermal transfer, dual-crop (640×640) spatial extraction into left and right viewports, video-level annotation in Label Studio, and full-rate 24 Hz frame extraction. Curated sequences are partitioned into 240-frame episodes under a parent-video grouped split protocol (TBD) with anti-adjacency snippet shuffling to eliminate cross-tile spatial leakage. At the edge inference stage, parallel lightweight YOLO11n backbones process incoming RGB and thermal frames at 24 Hz ($T_s \approx 41.7$ ms), integrated via soft disjunctive late fusion ($s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$) and evaluated across five lightweight temporal decision methods ($W = 5$ frames, 208 ms latency bound). Alert decisions are benchmarked across positive event detection, false alert mitigation, decision delay, and edge processing overhead savings.

false alarms on empty scenes, while transient confidence dips cause false dismissals.

To resolve these challenges without conflating model architecture with domain generalization, we freeze our study around two environments (arid desert and temperate forest) and two modalities (visible RGB and thermal infrared). We train two separate upstream YOLO11n detectors: one specialized on multi-environment RGB data and one specialized on multi-environment thermal data. Downstream, causal temporal post-processing filters the confidence stream across a causal sliding window of five frames (208 ms at a 24 Hz frame rate). We evaluate deterministic smoothing filters, order-statistic filters, temporal consensus tracking, and a learned Mamba selective state-space model (SSSM) [10], with each method operating under an optimal decision threshold farmed from validation video sequences.

The central research question is: *How do lightweight causal temporal decision methods compare in generating reliable and efficient alerts from multimodal RGB and thermal detections for autonomous aerial search and rescue?*

This work delivers five main contributions:

- 1) **Controlled 2×2 Video Corpus:** A dual-stream RGB and thermal benchmark across arid desert and temperate forest environments, structured into positive target scenarios, challenging hard-negative distractors (heated rocks, moving canopy shadows), and clear-negative scenes.
- 2) **Modality-Specific Detection Baselines:** A standardized edge detection framework deploying two separate YOLO11n models (2.6M parameters, 6.5 GFLOPs): one trained on multi-environment RGB sequences and one on multi-environment thermal sequences, using batch

size 16 and FP32 arithmetic for deterministic numerical stability.

- 3) **Multimodal Alert Decision Benchmark:** An empirical evaluation of five lightweight temporal decision techniques (raw thresholding, moving average, median filtering, history consensus, and Mamba-SSSM) across positive event detection, false alert mitigation, decision delay, and processing overhead under method-specific validation-optimized thresholds.
- 4) **Modality and Environment Sensitivity Analysis:** Systematic characterization of how temporal decision filters behave differently across RGB and thermal confidence streams and across disparate biomes.
- 5) **Operational Resource Impact Analysis:** Formal modeling of downstream benefits on robotic battery life and IoT telemetry bandwidth conservation.

The remainder of this paper is organized as follows: Section II reviews related literature in multimodal aerial detection, temporal post-processing, and state-space sequence models. Section III presents the proposed multimodal sensing, decision fusion, and causal temporal filtering methodology. Section IV outlines the controlled experimental design and dataset corpus. Section V discusses the comparative detection benchmarks, operational resource impacts, and experimental limitations (Section V-D), and Section VI concludes the paper.

II. RELATED WORK

A. Multispectral and Multimodal Person Detection

Multispectral detection combines complementary spectral bands to overcome illumination limits. Hwang et al. [4] introduced the KAIST multispectral benchmark for urban surveillance, proving that thermal radiation preserves human

contours when visible lighting degrades. Jia et al. [5] introduced LLVIP, showing significant low-light improvements by pairing infrared cues with optical streams. Architectures generally employ early, halfway, or late fusion. Liu et al. [11] demonstrated that halfway and late fusion strategies preserve modality-specific representations better than early channel concatenation. Li et al. [12] developed an illumination-aware weighting network, while Zhang et al. [13] proposed cross-modality interactive attention to suppress noisy features. Wilderness SAR presents localized degradations such as sudden canopy shade or solar ground heating. In this work, we adopt soft disjunctive late fusion via max-pooling, allowing each detector branch to run independently without cross-channel synchronization bottlenecks.

B. Lightweight Detectors for Aerial Search and Rescue

Deploying vision algorithms on UAVs requires balancing detection accuracy against SWaP-C limits. Rudol and Doherty [1] demonstrated thermal and visible person detection for aerial search missions. Silvagni et al. [2] highlighted that battery reserves strictly constrain UAV flight duration. Lygouras et al. [14] showed that complex neural ensembles exceed the compute envelope of onboard embedded units. Portmann et al. [6] identified that sun-warmed rocks and terrain clutter produce thermal crossover false alarms. Kyrkou and Theodoridis [3] addressed communication bottlenecks with EmergencyNet, demonstrating that erroneous aerial alerts congest wireless telemetry links. Single-stage object detectors have evolved from YOLO [7] through YOLOv7 [8], YOLOv9 [15], YOLOv10 [16], and YOLO11 [9]. We select YOLO11n as our edge detector, providing 2.6M parameters and 6.5 GFLOPs per forward pass.

C. Temporal Post-Processing and Sequential Decision Making

Applying frame-by-frame detectors to video streams induces high-frequency confidence chatter. In video object detection, Han et al. [17] proposed Sequence Non-Maximum Suppression (Seq-NMS) to link boxes across time, while Kang et al. [18] introduced Tubelet Convolutional Neural Networks (T-CNN) using tubelets. However, multi-frame tracking pipelines incur optical flow and graph optimization latency unsuitable for edge UAVs. In contrast, our approach operates directly on the scalar confidence stream. The foundation of sequential decision making dates to Wald’s sequential probability ratio test [19]. In radar signal processing, Richards [20] formulated sliding-window M -out-of- N binary integration to suppress transient pulses while bounding latency. Rabiner et al. [21] established the noise-rejection properties of running median and moving average filters. We benchmark five-frame history consensus, moving average, and median filtering under validation-optimized thresholds.

D. State-Space Sequence Models

Recurrent networks capture temporal dependencies but suffer from gradient degradation, while Transformers exhibit

TABLE I
SUMMARY OF KEY MATHEMATICAL NOTATIONS AND SYSTEM PARAMETERS

Symbol / Variable	Physical Description / Operational Definition
$I_{\text{RGB}}[n], I_{\text{Thermal}}[n]$	Synchronized visible RGB and thermal infrared video frames at discrete index n
$c_{\text{RGB}}[n], c_{\text{Thermal}}[n]$	Upstream YOLO11n detection confidence scores $\in [0.0, 1.0]$
$s[n]$	Fused confidence score via soft disjunctive late fusion $\in [0.0, 1.0]$
f_s, T_s	Operating timebase ($f_s = 24$ Hz, sampling interval $T_s \approx 41.7$ ms)
W	Sliding temporal post-processing window length ($W = 5$ frames, ≈ 208 ms)
$\mathcal{W}[n]$	Causal sliding window observation vector $\{s[n-4], \dots, s[n]\}$
$\mathcal{M}_m$	Causal post-processing method $m \in \{M1, \dots, M5\}$
τ_m^*	Optimal decision threshold for method m farmed on validation partition $\mathcal{D}_{\text{val}}$
$\mathcal{D}_{\text{val}}$	Disjoint validation video sequence partition used for threshold optimization
$\Delta\tau$	Threshold optimization sweep step size ($\Delta\tau = 0.02$, 46 candidate thresholds)
$y[n]$	Binary alarm decision asserted at frame n ($y[n] \in \{0, 1\}$)
d_{model}	Hidden projection dimension for Mamba-SSSM architecture ($d_{\text{model}} = 16$)
d_{state}	Recurrent hidden state dimension for Mamba-SSSM architecture ($d_{\text{state}} = 8$)
A, B_t, C_t	State transition, input control, and output projection matrices
Δt	Input-dependent selective discretization time-step parameter
h_t	Recurrent state tensor in Mamba-SSSM ($h_t \in \mathbb{R}^{d_{\text{model}} \times d_{\text{state}}}$)
u_t	Intermediate selective state projection representation
S_{pkt}	Uplink telemetry packet transmission size (1.2 kB standard, 45 kB thumbnail)
K_{FP}	Cumulative false positive alarm activation count on negative sequences
Ω_{wasted}	Cumulative wasted uplink telemetry bandwidth
P_{hover}	UAV hovering electrical power draw ($P_{\text{hover}} \approx 280$ W)
T_{loiter}	Target verification loiter maneuver duration (15–30 s)
E_{loiter}	Consumed battery energy per verification loiter maneuver
FASR	False Alarm Suppression Ratio relative to baseline raw thresholding M1
$T_{\text{RGB}}, T_{\text{Thermal}}$	Inference latency per frame of upstream YOLO11n-RGB and YOLO11n-Thermal
$T_{\text{fusion}}, T_{\text{postproc}}$	Latency per frame of decision fusion gate and causal post-processing filter

quadratic complexity with sequence length. Structured state-space models (SSMs) offer an efficient alternative. Gu et al. [22] introduced S4, modeling continuous-time state spaces with linear complexity. Gu and Dao [10] proposed Mamba, introducing data-dependent selection mechanisms and hardware-efficient parallel scans. Vision adaptations include Vision Mamba [23] and VMamba [24]. Dao and Gu [25] established structured state-space duality connecting recurrence and attention. Here, we adapt a pure-PyTorch selective SSM (Mamba-SSSM) as a causal temporal post-processor to filter environmental noise spikes.

III. METHODOLOGY

The proposed MMSAR framework provides an end-to-end pipeline combining lightweight dual-modality detection, soft disjunctive late fusion, and causal temporal post-processing (Fig. 1). Table I outlines the key mathematical notations and system parameters used throughout this work.

A. Video Stream Ingestion and Upstream Modality Detectors

Embedded UAV companion computers (e.g., NVIDIA Jetson Orin Nano, 7–15 W thermal design power (TDP)) deployed on autonomous search platforms require real-time execution matching the sensor stream. In this framework, synchronized video streams are ingested and processed at the native video capture rate of $f_s = 24$ Hz ($T_s = 1/24$ s ≈ 41.7 ms), providing continuous 24 FPS frame-level monitoring across harvested 640×640 pixel snippets.

To guarantee strictly causal, frame-synchronous execution without buffering latency or frame drops on embedded avionics, the end-to-end processing pipeline per frame must satisfy the real-time budget constraint (serial execution Eq. (1)):

$$T_{\text{RGB}} + T_{\text{Thermal}} + T_{\text{fusion}} + T_{\text{postproc}} \leq T_s \approx 41.7 \text{ ms}, \quad (1)$$

or under parallel multi-threaded branch execution across asynchronous edge cores (Eq. (2)):

$$\max(T_{\text{RGB}}, T_{\text{Thermal}}) + T_{\text{fusion}} + T_{\text{postproc}} \leq T_s \approx 41.7 \text{ ms}, \quad (2)$$

where T_{RGB} and T_{Thermal} represent the inference latencies of the respective YOLO11n backbones, T_{fusion} represents the scalar pairwise max-pooling latency, and T_{postproc} denotes the execution latency of post-processing filter $\mathcal{M}_m$. As detailed in Section III-C and Table II, both T_{fusion} ($O(1)$) and T_{postproc} (sub-millisecond across all methods M1–M5) are computationally negligible compared to deep feature extraction. Consequently, real-time feasibility hinges almost entirely on upstream backbone execution ($T_{\text{RGB}}, T_{\text{Thermal}}$: [TBD — pending on-device benchmarking]).

To isolate temporal post-processing dynamics from multimodal co-training artifacts, we train two distinct upstream YOLO11n models [9]:

- 1) **YOLO11n-RGB**: Trained on multi-environment RGB training sequences (desert and forest), outputting confidence $c_{\text{RGB}}[n] \in [0.0, 1.0]$.
- 2) **YOLO11n-Thermal**: Trained on multi-environment thermal training sequences (desert and forest), outputting confidence $c_{\text{Thermal}}[n] \in [0.0, 1.0]$.

Each detector possesses 2.6M parameters and 6.5 GFLOPs at 640×640 resolution. If no candidate exceeds non-maximum suppression (NMS), the frame confidence is assigned as 0.0.

B. Soft Disjunctive Late Fusion Gate

To prevent single-modality dropouts from causing target misses, we integrate per-frame detector outputs via soft disjunctive late fusion using max-pooling:

$$s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n]) \in [0.0, 1.0]. \quad (3)$$

The disjunctive formulation ensures that an unambiguous detection in either sensor branch propagates forward with full confidence. Furthermore, decision-level late fusion avoids expensive feature alignment layers, allowing each sensor branch to execute asynchronously on multicore edge processors. Algorithm 1 formalizes the complete per-frame causal alarm decision pipeline.

C. Causal Temporal Post-Processing

Direct thresholding of confidence streams yields high false-alarm rates due to isolated spikes. Downstream processing filters the confidence stream across a causal sliding window of length $W = 5$ frames (208 ms at $f_s = 24$ Hz):

$$\mathcal{W}[n] = \{s[n-4], s[n-3], s[n-2], s[n-1], s[n]\}. \quad (4)$$

Because mathematical filters exhibit distinct transfer functions, enforcing a uniform threshold penalizes performance. Each method $m \in \{\text{M1}, \dots, \text{M5}\}$ is assigned an individual optimal threshold τ_m^* farmed on the disjoint validation set $\mathcal{D}_{\text{val}}$:

$$\tau_m^* = \arg \max_{\tau \in [0.05, 0.95]} F_1(\tau; \mathcal{D}_{\text{val}}, \mathcal{M}_m), \quad (5)$$

with step $\Delta\tau = 0.02$ (yielding 46 candidate evaluation thresholds). Once optimized on $\mathcal{D}_{\text{val}}$, thresholds are permanently

Algorithm 1 MMSAR Causal Alarm Decision (per frame n)

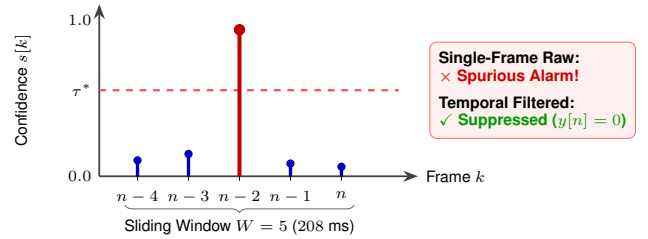
Require: Synchronized RGB frame $I_{\text{RGB}}[n]$, thermal frame

$I_{\text{Thermal}}[n]$

Ensure: Binary alarm $y[n] \in \{0, 1\}$

- 1: $c_{\text{RGB}}[n] \leftarrow \text{YOLO11n-RGB}(I_{\text{RGB}}[n])$
- 2: $c_{\text{Thermal}}[n] \leftarrow \text{YOLO11n-Thermal}(I_{\text{Thermal}}[n])$
- 3: $s[n] \leftarrow \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$ {Eq. (3)}
- 4: Update causal window $\mathcal{W}[n] \leftarrow \{s[n-4], \dots, s[n-1], s[n]\}$
- 5: $y[n] \leftarrow \mathcal{M}_m(\mathcal{W}[n]; \tau_m^*)$ {Method $m \in \{\text{M1}, \dots, \text{M5}\}$, Eqs. (6)–(14)}
- 6: **if** $y[n] = 1$ **then**
- 7: Emit telemetry packet (S_{pkt}) and trigger UAV loiter maneuver
- 8: **end if**
- 9: **return** $y[n]$

(a) Transient Noise Spike (Single-Frame Clutter)



(b) Sustained Target Signature (True SAR Victim)

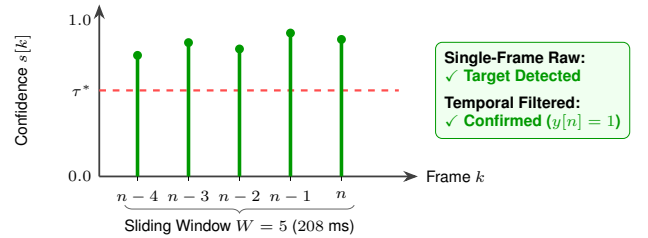


Fig. 2. Temporal signal behavior ($W = 5$ frames ≈ 208 ms at 24 Hz): (a) an isolated noise spike on a hard-negative scene triggers a spurious raw detection but is rejected by causal filtering ($y[n] = 0$), whereas (b) a sustained victim signature confirms target presence and asserts the alarm ($y[n] = 1$).

frozen prior to test set evaluation. The complete sweep trajectory $\{F_1(\tau; \mathcal{D}_{\text{val}}, \mathcal{M}_m)\}_{\tau=0.05}^{0.95}$ is logged and retained across all candidate thresholds to construct continuous operating curves (Fig. 3), rather than collapsing filter characterization to a single operating point.

We benchmark five causal post-processing methods:

- **M1 (Baseline Raw Thresholding)**: Memoryless frame thresholding asserted via Eq. (6):

$$y_{\text{raw}}[n] = \mathbb{I}(s[n] \geq \tau_{\text{raw}}^*). \quad (6)$$

Lacking temporal memory, it is vulnerable to transient distractor spikes.

[TODO: Full Validation Threshold Operating Curves for M1–M5]

Precision-Recall and F_1 -vs- τ curves across $\tau \in [0.05, 0.95]$ ($\Delta\tau = 0.02$, 46 candidate thresholds), retained rather than collapsed to a single point at τ_m^* .

Fig. 3. Precision-Recall (or F_1 -vs- τ) operating curves for M1–M5 across the full validation threshold sweep, retained rather than collapsed to a single point at τ_m^* .

- **M2 (Five-Frame Moving Average):** Causal arithmetic mean, computed via Eq. (7) and asserted via Eq. (8):

$$\tilde{s}_{\text{MA}}[n] = \frac{1}{W} \sum_{k=0}^{W-1} s[n-k], \quad (7)$$

$$y_{\text{MA}}[n] = \mathbb{I}(\tilde{s}_{\text{MA}}[n] \geq \tau_{\text{MA}}^*). \quad (8)$$

A single-frame spike of 1.0 is attenuated to 0.20. However, moving averages attenuate rising step edges, requiring a lower optimal threshold τ_{MA}^* to preserve recall.

- **M3 (Five-Frame Median Filter):** Order-statistic filter, computed via Eq. (9) and asserted via Eq. (10):

$$\tilde{s}_{\text{med}}[n] = \text{median}(s[n], \dots, s[n-4]), \quad (9)$$

$$y_{\text{med}}[n] = \mathbb{I}(\tilde{s}_{\text{med}}[n] \geq \tau_{\text{med}}^*). \quad (10)$$

It rejects impulses spanning fewer than $\lfloor W/2 \rfloor + 1 = 3$ frames while preserving step edges during target entry.

- **M4 (Five-Frame History Consensus):** Sliding-window M -out-of- N integration ($M = 3, W = 5$) asserted via Eq. (11):

$$y_{\text{hist}}[n] = \mathbb{I}\left(\sum_{k=0}^{W-1} \mathbb{I}(s[n-k] \geq \tau_{\text{hist}}^*) \geq 3\right). \quad (11)$$

An alarm is confirmed only if at least 3 of the past 5 frames independently exceed candidate threshold τ_{hist}^* .

- **M5 (Learned Mamba-SSSM):** A pure-PyTorch selective state-space model. Scalar confidence $s[n]$ is projected to hidden dimension $d_{\text{model}} = 16$ via the Sigmoid Linear Unit (SiLU) activation: $x_t = \text{SiLU}(W_{\text{in}}s[n] + b_{\text{in}})$. Linear projections compute time-varying step size $\Delta_t = \text{softplus}(W_{\Delta}x_t + b_{\Delta})$ and transition matrices $B_t = W_Bx_t$, $C_t = W_Cx_t \in \mathbb{R}^{d_{\text{model}} \times d_{\text{state}}}$, with state dimension $d_{\text{state}} = 8$. Diagonal transition matrix $A = -\exp(A_{\text{log}})$ enforces stability. Under zero-order hold (ZOH) discretization (Eq. (12)), recurrent state updates (Eq. (13)), and alarm assertion (Eq. (14)) proceed as:

$$\bar{A}_t = \exp(\Delta_t \cdot A), \quad \bar{B}_t = (\Delta_t \cdot B_t) \odot x_t, \quad (12)$$

TABLE II
THEORETICAL PER-FRAME COMPLEXITY AND STATE MEMORY
(INDEPENDENT OF HARDWARE)

Method	Time / Frame	State Memory
M1: Raw Thresholding	$\mathcal{O}(1)$	0 B (memoryless)
M2: Moving Average	$\mathcal{O}(1)$ amortized*	$5 \times 4 \text{ B} = 20 \text{ B}$
M3: Median Filter	$\mathcal{O}(W)$ (sorted insert)	$5 \times 4 \text{ B} = 20 \text{ B}$
M4: History Consensus	$\mathcal{O}(W)$	5 B (packed booleans)
M5: Mamba-SSSM	$\mathcal{O}(d_{\text{model}} \cdot d_{\text{state}})$ (753 params) [†]	$16 \times 8 \times 4 \text{ B} = 512 \text{ B}$

*With a circular buffer / running sum; naive resummation is $\mathcal{O}(W)$.

[†]Parameter count verified programmatically on PyTorch implementation (753 trainable parameters).

$$h_t = \bar{A}_t \odot h_{t-1} + \bar{B}_t, \quad u_t = \sum_{j=1}^{d_{\text{state}}} h_{t,j} \odot C_{t,j}, \quad (13)$$

$$y_{\text{Mamba}}[n] = \mathbb{I}(\sigma(W_{\text{out}}u_t + b_{\text{out}}) \geq \tau_{\text{Mamba}}^*). \quad (14)$$

Mamba-SSSM maintains only h_t in memory ($16 \times 8 \times 4 \text{ bytes} = 512 \text{ bytes}$), achieving strict $\mathcal{O}(1)$ complexity per frame.

Table II contrasts the theoretical computational complexity and state memory footprint of methods M1–M5.

To establish statistical confidence without making parametric distributional assumptions over temporally correlated video frames, we employ non-parametric sequence-level bootstrapping. Specifically, we resample the 12 held-out test clips with replacement across $B = 1000$ bootstrap iterations. For each bootstrap replicate, detection predictions are evaluated to generate an empirical distribution of F_1 -scores, from which the two-sided 95% percentile confidence intervals (2.5th and 97.5th percentiles) are extracted and reported in Table VI.

Fig. 2 depicts the operational contrast between transient noise rejection and sustained target confirmation.

D. Downstream Operational Metrics

In aerial search, frame-level false positives incur mission-level costs:

- 1) **Uplink Telemetry Bandwidth:** Confirmed alarms ($y[n] = 1$) trigger emergency telemetry packets of size S_{pkt} (1.2 kB telemetry, or 45 kB with thumbnail) over satellite or cellular IoT links [3]. Wasted uplink bandwidth across K_{FP} false alarms is:

$$\Omega_{\text{wasted}} = K_{\text{FP}} \cdot S_{\text{pkt}} \quad [\text{bytes}]. \quad (15)$$

- 2) **Drone Flight Battery Drain:** Each alarm requires the UAV to loiter over coordinates for verification duration T_{loiter} (15–30 s). With hovering power $P_{\text{hover}} \approx 280 \text{ W}$, the energy expended is:

$$E_{\text{loiter}} = P_{\text{hover}} \cdot T_{\text{loiter}} \quad [\text{Joules}]. \quad (16)$$

We evaluate the False Alarm Suppression Ratio (FASR, Eq. (17)):

$$\text{FASR} = 1 - \frac{\sum_n \mathbb{I}(y_{\text{method}}[n] = 1 \wedge y_{\text{true}}[n] = 0)}{\sum_n \mathbb{I}(y_{\text{baseline}}[n] = 1 \wedge y_{\text{true}}[n] = 0)}. \quad (17)$$

TABLE III
CONTROLLED 2×2 EXPERIMENTAL DESIGN

	Arid Desert	Temperate Forest
RGB (YOLO11n-RGB)	30 clips (10/10/10)	30 clips (10/10/10)
Thermal (YOLO11n-Thermal)	30 clips (10/10/10)	30 clips (10/10/10)
10/10/10 = positive / hard-negative distractor / clear-negative clips.		

IV. EXPERIMENTAL SETUP AND DATASET

A. Synthetic Multimodal Dataset Specifications

Wilderness SAR dual-spectrum datasets remain scarce. As summarized in Table III, the controlled 2×2 experimental design balances modality and environment across identical scene categorizations.

To establish a controlled benchmark without overextending experimental conditions, we freeze our dataset around two primary wilderness biomes:

- 1) **Arid Desert:** High ambient temperatures, rocky outcroppings, sparse shrubs, and midday thermal crossover where ground rock temperatures equilibrate with human skin radiation.
- 2) **Temperate Forest:** Dense tree canopies, dappled ground shadows, intermittent foliage occlusions, and wildlife distractors.

Each environment is structured into three operational scenario categories: positive clips depicting human targets (walking, crawling, partially occluded), hard-negative distractor clips containing sun-heated boulders, moving canopy shadows, and animal clutter without persons, and clear-negative clips capturing empty wilderness terrain.

Raw aerial video captures are recorded in 1280×720 widescreen format at native 24 FPS (10.0 s duration, 240 frames). To preserve optical ground sampling distance (GSD) without scaling distortion or blurring sub-20-pixel targets, we execute a dataset $2\times$ dual-crop harvesting procedure: from each 1280×720 video, two disjoint 640×640 square snippets are harvested (left tile spanning $x \in [0, 640]$ and right tile spanning $x \in [640, 1280]$, both centered vertically at offset $y = 40$ pixels). This produces 30 distinct 10-second 24 FPS video sequences per environment (10 positive, 10 hard-negative distractor, and 10 clear-negative clips). To support web-based temporal scrubbing in Label Studio, raw MP4 files are remuxed with `-movflags +faststart`, relocating the `moov` atom to offset 36 for HTTP byte-range streaming. Annotators placed keyframe bounding boxes at target motion inflection points with linear track interpolation. Negative frames are explicitly recorded with empty 0-byte label files to penalize background false positives during detector training.

B. Sequence Partitioning and Detector Training

Partitioning is conducted strictly at the sequence level to eliminate temporal leakage. Clips within each environment and scenario category are divided into a 60/20/20 split: 60% for training (6 clips per scenario), 20% for validation (2 clips per scenario), and 20% for testing (2 clips per scenario).

TABLE IV
DATASET PARTITIONING AND DETECTOR TRAINING PARAMETERS

Parameter / Specification	Configuration Value
Environments	Arid Desert, Temperate Forest
Visual Modalities	Visible RGB, Long-Wave Thermal Infrared (TIR)
Raw Video Format	24 FPS (1280×720 pixels, 10.0 s duration)
Dual-Crop Harvesting	$2\times$ square snippets (640×640 , left & right tiles, $y = 40$)
Harvested Sequence Count	30 clips per env. (10 pos., 10 hard-neg., 10 clear-neg.)
Operating Timebase	$f_s = 24$ Hz ($T_s \approx 41.7$ ms, 240 frames / clip)
Sequence Split (Disjoint)	Train: 60%, Validation: 20%, Test: 20%
Upstream Detectors	Two YOLO11n Models (YOLO11n-RGB, YOLO11n-Thermal)
Detector Complexity	2.6M Parameters, 6.5 GFLOPs (640×640 resolution)
Arithmetic Precision	Full Single-Precision (FP32, <code>amp=False</code>)
Batch Size / Epochs	Batch 16 / 100 Epochs (no early stopping)
Optimizer	Stochastic Gradient Descent (SGD, momentum 0.937, weight decay 0.0005)
Learning Rate Schedule	Initial $lr_0 = 0.01$ with cosine decay
Sliding Window (W)	5 frames (208 ms causal latency bound)

TABLE V
FRAME-LEVEL UPSTREAM DETECTION PERFORMANCE ACROSS MODALITIES AND ENVIRONMENTS (TEST SPLIT, $\tau = 0.50$)

Configuration / Condition	Precision	Recall	F1-Score
YOLO11n-RGB (Arid Desert)	[TBD]	[TBD]	[TBD]
YOLO11n-RGB (Temperate Forest)	[TBD]	[TBD]	[TBD]
YOLO11n-Thermal (Arid Desert)	[TBD]	[TBD]	[TBD]
YOLO11n-Thermal (Temperate Forest)	[TBD]	[TBD]	[TBD]
Late Fusion Gate (Desert)	[TBD]	[TBD]	[TBD]
Late Fusion Gate (Forest)	[TBD]	[TBD]	[TBD]

Frames within each clip are preserved in exact sequential order ($0 \rightarrow 239$) to support sequential state-space modeling. Table IV summarizes the experimental parameters.

V. RESULTS AND DISCUSSION

A. Upstream Detection Across Modalities and Environments

We evaluate the upstream frame-level detection performance of the two YOLO11n models on the test split. Table V summarizes precision, recall, and F1-score across modalities and environments under standard thresholding ($\tau = 0.50$).

Qualitatively, the individual streams exhibit complementary trade-offs:

- 1) Visible RGB detection delivers sharp localization in open illuminated desert scenes, but experiences target dropouts under dense tree foliage and dark canopy shadows in forest scenes.
- 2) Thermal infrared detection maintains target visibility through shadows and light foliage, but suffers from elevated false positive spikes on solar-heated rocks and sand during midday desert passes.
- 3) Soft disjunctive late fusion combines both streams via max-pooling, increasing target recall by ensuring that detections in either spectrum propagate forward.

However, because max-pooling propagates the highest score, raw fusion also passes transient false alarm spikes from both sensors, necessitating causal temporal filtering.

B. Comparative Evaluation of Causal Post-Processing Methods

The five causal post-processing methods are evaluated on the common confidence streams across the test split. Each method executes under its optimal decision threshold τ_m^*

farmed on validation sequences $\mathcal{D}_{\text{val}}$ via Eq. (5). Table VI reports the comparative benchmark.

The post-processing methods demonstrate distinct filtering characteristics:

- **M1 (Baseline Raw):** Lacking temporal memory, memoryless thresholding allows isolated single-frame distractor spikes to trigger alarms, resulting in zero false-alarm suppression on negative sequences.
- **M2 (Moving Average):** Causal arithmetic averaging attenuates isolated single-frame spikes by factor $1/W = 0.20$. Because averaging attenuates target onset edges, the optimal threshold τ_{MA}^* adapts downward to balance recall.
- **M3 (Median Filter):** The median operator eliminates noise bursts spanning fewer than 3 frames without attenuating sustained step edges once targets enter view.
- **M4 (History Consensus):** Requiring at least 3 of 5 frames to meet threshold τ_{hist}^* provides discrete temporal confirmation, suppressing transient clutter.
- **M5 (Mamba-SSSM):** The selective state-space model dynamically adjusts its recurrent state. Transient spikes decay rapidly under dissipative state dynamics, whereas sustained target motion drives the hidden state into the alarm regime. Its pure-PyTorch implementation executes with sub-millisecond latency on edge hardware.

C. Environmental Robustness and Downstream Operational Impact

Across environmental conditions, distinct operational benefits emerge:

- 1) **Arid Desert:** Sun-heated boulders and sand create thermal crossover false spikes. Temporal post-processing filters these isolated spikes, preventing spurious triggers.
- 2) **Temperate Forest:** Canopy occlusions cause brief 1-to-2 frame dropouts in the RGB stream. Soft disjunctive fusion combined with temporal state memory bridges these gaps to preserve target tracks.

Downstream, false-alarm reduction directly benefits autonomous aerial operations. On bandwidth-limited IoT links (e.g., satellite links operating at 2.4–10 kbps), suppressing spurious alarms prevents telemetry queue congestion and packet loss [3]. Furthermore, because each alarm mandates a 15–30 s UAV loiter maneuver consuming $P_{\text{hover}} \approx 280$ W, eliminating dozens of false alarms conserves tens of kilojoules of battery reserves per sortie, directly expanding operational flight endurance and search coverage.

Table VII models the environment-stratified operational savings achieved by each causal post-processing filter relative to baseline raw thresholding (M1). For each method m and environment $e \in \{\text{Desert, Forest}\}$, the number of false positive alarms avoided is quantified as $\Delta\text{FP}_m(e) = \text{FP}_{\text{baseline}}(e) - \text{FP}_m(e)$. Feeding $\Delta\text{FP}_m(e)$ into Eq. (15), the total uplink telemetry bandwidth preserved is $\Delta\Omega_m(e) = \Delta\text{FP}_m(e) \cdot S_{\text{pkt}}$, evaluated for standard telemetry packets ($S_{\text{pkt}} = 1.2$ kB) and visual thumbnail verification packets ($S_{\text{pkt}} = 45$ kB). Using Eq. (16), the cumulative battery energy conserved by averting unneeded loiter confirmation maneuvers is $\Delta E_m(e) =$

$\Delta\text{FP}_m(e) \cdot P_{\text{hover}} \cdot T_{\text{loiter}}$, where $P_{\text{hover}} \approx 280$ W and loiter verification duration $T_{\text{loiter}} \in [15, 30]$ s. The corresponding extension in search sortie endurance is projected directly as $\Delta t_{\text{flight}} = \Delta E_m(e) / P_{\text{hover}} = \Delta\text{FP}_m(e) \cdot T_{\text{loiter}}$.

D. Limitations

While the comparative framework provides systematic insights into causal temporal post-processing for aerial SAR, several operational and experimental constraints bound the scope of the present findings:

- 1) **Sample Size of Test Split:** The evaluation is conducted across 12 held-out test video sequences (2 clips per scenario per environment, totaling 2,880 test frames at 24 Hz). Although sequence-level partitioning strictly prevents temporal data leakage, broader multi-hour flight campaigns are required for comprehensive statistical characterization.
- 2) **Biome Diversity:** The study is constrained to two representative wilderness biomes (arid desert and temperate forest). Maritime, alpine snowpack, wetland, and urban-wildland interface environments introduce distinct thermal-optical scattering profiles not captured in this corpus.
- 3) **Flight Regime and Dynamics:** Video sequences model a constrained aerial flight regime (30–50 m above ground level at flight speeds of 8–15 m/s). High-speed turbulent sorties or low-altitude hover-scan maneuvers may alter apparent target ground sampling distance and jitter profiles.
- 4) **Atmospheric and Weather Phenomena:** All clips reflect nominal daytime operating conditions; severe weather degradations including dense advection fog, heavy precipitation, and blowing dust are unmodeled.
- 5) **Single-Target Assumption:** The current formulation focuses on individual person search scenarios and does not incorporate multi-target identity disambiguation or crowded group spatial tracking.
- 6) **Absence of Physical Hardware-in-the-Loop Validation:** Algorithmic latencies and memory bounds are derived theoretically and benchmarked in software; physical deployment aboard an integrated flight avionics stack (e.g., NVIDIA Jetson companion computer interfacing with a Pixhawk flight controller) remains unvalidated in this pass.

VI. CONCLUSION

This paper investigated the integration of lightweight multi-modal person detection with causal temporal post-processing for UAV wilderness search and rescue across two environments (arid desert and temperate forest) and two visual modalities (visible RGB and thermal infrared). We deployed two separate YOLO11n models (2.6M parameters, 6.5 GFLOPs at 640×640) trained with batch size 16 in FP32 precision as our edge detection baselines, combined via soft disjunctive late fusion. Downstream, five causal post-processing methods

TABLE VI

COMPARATIVE BENCHMARK OF CAUSAL TEMPORAL POST-PROCESSING METHODS UNDER VALIDATION-OPTIMIZED DECISION THRESHOLDS ($W = 5$ FRAMES / 208 MS AT 24 HZ)

Method	Val τ^*	Precision	Recall	F1-Score	F1 (95% CI)	Pos. Recall (%)	Hard-Neg FP	Clear-Neg FP	FASR (%)	Latency / Frame
M1: Baseline Raw Thresholding	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M2: Five-Frame Moving Average	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M3: Five-Frame Median Filter	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M4: Five-Frame History Consensus	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M5: Learned Mamba-SSSM	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

TABLE VII

ENVIRONMENT-STRATIFIED FALSE-ALARM SUPPRESSION AND PROJECTED OPERATIONAL SAVINGS

Method	Environment	FASR (%)	FP Avoided vs. M1	Bandwidth Saved (kB)	Battery Saved (kJ)	Est. Added Flight Time (min)
M2: Moving Average	Arid Desert	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M2: Moving Average	Temperate Forest	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M3: Median Filter	Arid Desert	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M3: Median Filter	Temperate Forest	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M4: History Consensus	Arid Desert	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M4: History Consensus	Temperate Forest	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M5: Mamba-SSSM	Arid Desert	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M5: Mamba-SSSM	Temperate Forest	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

operating across a five-frame (208 ms at 24 Hz) sliding window were benchmarked under validation-optimized decision thresholds. Causal temporal post-processing effectively filters transient environmental noise spikes across modalities and terrains while maintaining rapid target confirmation, conserving telemetry bandwidth and onboard battery reserves.

Future work will focus on hardware-in-the-loop flight testing on NVIDIA Jetson companion computers aboard physical UAVs, spatio-temporal multi-target tracking, and domain adaptation to unmodeled weather conditions such as fog and rain.

DATA AND CODE AVAILABILITY

To ensure reproducibility and facilitate community benchmarking in wilderness search and rescue, the MMSAR simulation framework, dual-crop snippet harvesting scripts, and pure-PyTorch causal post-processing implementations are maintained at [PLACEHOLDER: code repository URL, or “available upon request from the corresponding author”]. Synthetic multimodal video clips and associated bounding-box annotation manifests are accessible at [PLACEHOLDER: dataset repository URL or DOI]. Pre-trained YOLO11n-RGB and YOLO11n-Thermal model weights are archived under [PLACEHOLDER: model weight repository URL].

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REFERENCES

- [1] P. Rudol and P. Doherty, “Human body detection and geolocation for UAV search and rescue operations using color and thermal imagery,” in *Proceedings of the IEEE Aerospace Conference*, 2008, pp. 1–8.
- [2] R. Silvagni, A. Tonoli, E. Zenerino, and M. Chiaberge, “Multipurpose UAV for search and rescue operations in mountain avalanche events: Design considerations,” *Geomatics, Natural Hazards and Risk*, vol. 8, no. 1, pp. 18–33, 2016.
- [3] C. Kyrkou and T. Theodoridis, “EmergencyNet: Efficient aerial image classification for drone-based emergency monitoring using atrous convolutional feature fusion,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 13, pp. 1687–1699, 2020.
- [4] S. Hwang, J. Park, N. Kim, Y. Choi, and I. S. Kweon, “Multispectral pedestrian detection: Benchmark dataset and baseline,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 1037–1045.
- [5] X. Jia, C. Zhu, M. Li, W. Tang, and W. Zhou, “LLVIP: A visible-infrared paired dataset for low-light vision,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 2021, pp. 3489–3497.
- [6] D. Portmann, S. Lynen, M. Chli, and R. Siegwart, “People detection and tracking from aerial thermal views,” in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2014, pp. 1794–1800.
- [7] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You only look once: Unified, real-time object detection,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.
- [8] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, “YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 7464–7475.
- [9] G. Jocher and J. Qiu, “Ultralytics YOLO11,” 2024. [Online]. Available: <https://github.com/ultralytics/ultralytics>
- [10] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” in *Proceedings of the First Conference on Language Modeling (COLM)*, 2024, also arXiv:2312.00752.
- [11] J. Liu, S. Zhang, S. Wang, and D. N. Metaxas, “Multispectral deep neural networks for pedestrian detection,” in *Proceedings of the British Machine Vision Conference (BMVC)*, 2016, pp. 73.1–73.13.
- [12] C. Li, D. Song, R. Tong, and M. Tang, “Illumination-aware faster R-CNN for robust multispectral pedestrian detection,” *Pattern Recognition*, vol. 85, pp. 161–171, 2019.
- [13] L. Zhang, Z. Liu, S. Zhang, X. Yang, H. Qiao, K. Huang, and A. Husain, “Cross-modality interactive attention network for multispectral pedestrian detection,” *Information Fusion*, vol. 50, pp. 20–29, 2019.
- [14] V. Lygouras, N. Santavas, A. Taitzoglou, K. Tarchanidis, A. Mitropoulos, and A. Gasteratos, “Unsupervised human detection with an embedded vision system on a fully autonomous UAV for search and rescue operations,” *Sensors*, vol. 19, no. 16, p. 3542, 2019.
- [15] C.-Y. Wang, I.-H. Yeh, and H.-Y. M. Liao, “YOLOv9: Learning what

- you want to learn using programmable gradient information,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024, pp. 1–18, also arXiv:2402.13616.
- [16] A. Wang, H. Chen, L. Liu, K. Chen, Z. Lin, J. Han, and G. Ding, “YOLOv10: Real-time end-to-end object detection,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 37, 2024, pp. 107 982–108 006, also arXiv:2405.14458.
 - [17] W. Han, P. Khorrani, T. L. Paine, P. Ramachandran, M. Babaeizadeh, H. Shi, J. Li, S. Yan, and T. S. Huang, “Seq-NMS for video object detection,” *arXiv preprint arXiv:1602.08465*, 2016.
 - [18] K. Kang, H. Li, T. Xiao, W. Ouyang, J. Yan, X. Liu, and X. Wang, “T-CNN: Tubelets with convolutional neural networks for object detection from videos,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 28, no. 10, pp. 2896–2907, 2018.
 - [19] A. Wald, “Sequential tests of statistical hypotheses,” *The Annals of Mathematical Statistics*, vol. 16, no. 2, pp. 117–186, 1945.
 - [20] M. A. Richards, *Fundamentals of Radar Signal Processing*, 2nd ed. New York, NY, USA: McGraw-Hill Education, 2014.
 - [21] L. R. Rabiner, M. R. Sambur, and C. E. Schmidt, “Applications of a nonlinear smoothing algorithm to speech processing,” *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 23, no. 6, pp. 552–557, 1975.
 - [22] A. Gu, K. Goel, and C. Ré, “Efficiently modeling long sequences with structured state spaces,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2022. [Online]. Available: <https://openreview.net/forum?id=uYLFoz1v1AC>
 - [23] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, “Vision mamba: Efficient visual sequence modeling with bidirectional state spaces,” in *Proceedings of the International Conference on Machine Learning (ICML)*, 2024, pp. 62 429–62 441, also arXiv:2401.09417.
 - [24] Y. Liu, Y. Tian, Y. Zhao, H. Yu, L. Xie, Y. Wang, Q. Ye, and Y. Liu, “VMamba: Visual state space model,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 37, 2024, pp. 103 031–103 063, also arXiv:2401.10166.
 - [25] T. Dao and A. Gu, “Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality,” in *Proceedings of the International Conference on Machine Learning (ICML)*, 2024, pp. 10 041–10 071, also arXiv:2405.21060.